University Health Network

Toronto General Hospital Toronto Western Hospital Princess Margaret Hospital

ACUTE TRANSFUSION REACTION

*as per Clinical Policy 3.130.005

For all unwanted signs/symptoms occurring within 24 hours of transfusion:

1. STOP transfusion
2. Maintain IV access with saline at 50cc/hr using new IV set
3. Check vital signs
4. Re check name and MRN of patient with product label
5. Notify patient's physician
6. Notify blood bank

Symptom	Risk Stratification	Possible Etiology	Order Transfusion Reaction in EPR?	Other Investigations Recommended?*	Can Transfusion be Resumed?	Send Copy of Mountsheet to Blood Bank
Fever (≥38ºC and increase of at least 1ºC from baseline) and/or **Chills/Rigors**	**Low risk** 38ºC to <39ºC and **no** other Abnormal signs or symptoms (eg., hypotension, rigors)	Febrile non-hemolytic transfusion reaction	Not Required	No	If patient remains stable, product still viable and attending MD approves, transfusion may be resumed cautiously at 50 ml/hr	Yes
	High risk <39ºC but with any other abnormal sign or symptoms **OR** ≥39ºC	Acute hemolytic transfusion reaction Bacterial contamination	Yes	Blood cultures x 2, aPTT, INR, fibrinogen, CBC, electrolytes, creatinine, bilirubin, LDH, haptoglobin and urinalysis	NO: return unused product to blood bank, plus empty bags/ bottles of products transfused in previous 4 hours (if available)	Yes
Rash (hives/urticaria/itching)	**Minor** < 2/3 of body and **no** other symptoms	Minor allergic transfusion reaction	Not Required	No	If patient remains stable, product still viable and attending MD approves, transfusion may be resumed cautiously at 50 ml/hr	Yes
	Major >2/3 of body and **no** other symptoms	Major allergic transfusion reaction	Not Required	No	NO: return unused product to blood bank	Yes
	Severe Accompanied by other abormal signs/symptoms (eg., chest pain, respiratory symptoms, facial edema, hypotension)	Anaphylactic transfusion reaction	Yes	*If accompanied by hypoxia:* CXR, arterial blood gas *If accompanied by hypotension*: Blood cultures x 2, aPTT, INR, fibrinogen, CBC, electrolytes, creatinine, bilirubin, LDH, haptoglobin and urinalysis	NO: return unused product to blood bank	Yes
Dyspnea (increased work of breathing, or SpO_2% decreased at least 5% from baseline)		Anaphylactic transfusion reaction Transfusion-associated circulatory overload Transfusion-related acute lung injury	Yes	Chest X-Ray, arterial blood gas *If accompanied by hypotension*: Blood cultures x 2, aPTT, INR, fibrinogen, CBC, electrolytes, creatinine, bilirubin, LDH, haptoglobin and urinalysis	NO: return unused product to blood bank	Yes

UHN Guidelines for Transfusion of Blood Components - Clinical Policy 3.130.001

Red Blood Cells

- 1 unit = 240 to 340 mL
- Infuse over 30 minutes to 4 hours
- Tubing: Y type tubing with in line blood filter 170-210 microns
- Indicated for augmentation of oxygen carrying capacity of blood; 1 unit will increase hemoglobin level of average-sized, non-bleeding adult by 10 g per L
- Indications as listed do not apply to patients with hemoglobinopathy (thalassemia, sickle cell disease) or autoimmune hemolytic anemia, who should only be transfused RBCs with a hematologist order and/or on-call transfusion medicine specialist review

Bleeding Patient

Clinical Setting	Recommendation	Dose
Low risk patient	Maintain hemoglobin 70 to 100 g per L during active bleeding	As per rate of blood loss
Impaired pulmonary function Impaired oxygen consumption Anesthesia Atherosclerosis	Maintain hemoglobin 80 to 100 g per L during active bleeding	As per rate of blood loss
Acute MI Unstable angina	Maintain hemoglobin greater than 100 g per L during active bleeding	As per rate of blood loss

Non-bleeding Patient

Hemoglobin Level	Recommendation	Dose
Less than 60 g per L	Transfusion highly recommended	1 to 2 units
Less than 70 g per L	Transfusion likely appropriate	1 unit
70 to 100 g per L	Transfusion likely appropriate if there are signs or symptoms of impaired oxygen delivery	1 unit
Greater than 100 g per L	Transfusion likely inappropriate	None

Plasma

- 1 single-donor unit = approx 100 to 250 mL frozen within 8 hours (fresh frozen) or 24 hours (frozen) of collection. 1 apheresis unit = approx 200 to 600 mL, frozen within 8 hours of collection. May be considered therapeutically equivalent to 2 single-donor units
- Infuse over 30 minutes to 2 hours (maximum 4 hours)
- Tubing: Y type tubing with in line blood filter 170-210 microns
- Indicated for improved hemostasis in patients with deficiencies of coagulation factors; a 10 to 15 mL per kg dose will increase all coagulation factor levels by approximately 20 to 30%
- Patients with coagulation factor inhibitors (e.g., due to amyloidosis or lupus anticoagulants) should only be transfused plasma with a hematologist order

Bleeding Patient

Clinical Setting	Recommendation	Dose
Therapy with vitamin K antagonist	Transfuse plasma if INR greater than 1.5	3 to 4 units **or** consider prothrombin complex concentrate as alternative; administer 10 mg IV vitamin K to prevent rebound
Liver disease Dilutional coagulopathy DIC	Transfuse plasma if INR greater than 1.5	3 to 4 units
Massive transfusion with microvascular bleeding, unable to wait for laboratory results	Transfuse plasma empirically	3 to 4 units

Non-Bleeding Patient

Clinical Setting	Recommendation	Dose
Therapy with vitamin K antagonist	Transfuse plasma if INR greater than 1.5 **and** surgical procedure within 12 hours which cannot be safely delayed	3 to 4 units or consider prothrombin complex concentrate as alternative; administer 1 to 2 vitamin K IV or PO to prevent rebound
Liver disease Dilutional coagulopathy DIC	Transfuse plasma if INR greater than 1.5 immediately prior to surgical procedure	3 to 4 units

Platelets

- 1 buffy coat pool (4 donors' whole blood platelet fractions) = at least 240 x 10^9 platelets, suspended in 300 mL of single donor plasma. 1 apheresis unit (1 donor) = at least 300 x 10^9 platelets suspended in 400 mL of the same donor's plasma
- Infuse over 30 to 60 minutes (maximum 4 hours)
- Tubing: Y type tubing with in line blood filter 170-210 microns
- Indicated for improved hemostasis in patient with impaired platelet function; 1 pool or one apheresis unit will increase platelet count of average-sized non-bleeding adult by 40 x 10^9 per L
- Patients with refractory thrombocytopenia (eg., suspected anti-HLA or -HPA antibodies) should only be transfused platelets with a hematologist order

Bleeding Patient

Clinical Setting	Recommendation	Dose
Immune thrombocytopenia	Transfuse platelets only if life- or limb-threatening bleeding	1 pool or apheresis unit
Non-immune thrombocytopenia	Keep platelet count greater than 50 x 10^9 per L for most patients, greater than 100 x 10^9 per L for patients with CNS bleeding	1 pool or apheresis unit
Platelet dysfunction (e.g., following recent ingestion of ASA or other antiplatelet agent; post-cardiopulmonary bypass)	Transfuse platelets empirically (regardless of platelet count) to maximum of 2 doses	1 pool or apheresis unit

Non-bleeding Patient

Platelet count	Recommendation	Dose
Less than 10 x 10^9 per L	Transfuse platelets if non-immune thrombocytopenia	1 pool or apheresis unit
Less than 20 x 10^9 per L	Transfuse platelets if non-immune thrombocytopenia in setting of temp ≥ 38.5ºC or coagulopathy, or prior to minor surgical procedure	1 pool or apheresis unit
Less than 50 x 10^9 per L	Transfuse platelets immediately prior to or following any surgical procedure that places patient at significant risk from bleeding (e.g., liver resection)	1 pool or apheresis unit
Less than 100 x 10^9 per L	Transfuse platelets immediately prior to or following major neurosurgical procedure	1 pool or apheresis unit

Cryoprecipitate

- 1 unit = approx 5 to 15 mL derived by partially thawing a unit of frozen plasma
- Infusion time less than 30 minutes (maximum 4 hours)
- Tubing: Y type tubing with in line blood filter 170-210 microns
- Indicated for improved hemostasis in patients with deficiencies of fibrinogen; each unit contains approximately 150 mg of fibrinogen
- Note that patients with multiple coagulation factor deficiencies will likely not benefit from cryoprecipitate; 4 units of plasma contain approximately the same quantity of fibrinogen as 8 units of cryoprecipitate

Clinical Setting	Recommendation	Dose
Microvascular bleeding	Transfuse plasma if fibrinogen < 1.0 g/L	8-10 units